
QEMU, KVM (Kernel Virtual Machine) audio setup - or how to get pulse audio working in kvm guest and host simultaneously

*Virtual machine audio setup – or how to get pulse audio working
(<https://mathiashueber.com/virtual-machine-audio-setup-get-pulse-audio-working/>)*

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In this post, I compare several options of how to get the sound output from an KVM virtual machine guest, back into the host. I will compare spice server, pulse audio and a hardware solution.

In any case – harmonize sample rates of host and guest system

A common issue with sound passthrough between guest and host in Linux KVM is no sound at all or lagging or chopped up sound playback In order to fix this make sure the host and the guest use the same default playback sample rate.

Setting default sample rate & user for audio in Linux

* this should fix no audio/ no audio sharing simultaneously between guest & host KVM ↓↓

#Open /etc/libvirt/qemu.conf

\$ sudo pluma /etc/libvirt/qemu.conf

#Search & uncomment the line: user = "example"

```
user = "example"
```

#Replace "example " with your username "<username>"

from: user = "example"

to: user = "<your username>"

#Restart the pulse audio service

\$ pulseaudio -k or **\$** killall pulseaudio

and reboot

with audio now working in Linux KVM qemu guests and Linux host simultaneously....
The instructions on the next pages may be considered somewhat auxiliary/ for tweaking

```
#open /etc/pulse/daemon.conf
```

```
$ sudo pluma /etc/pulse/daemon.conf
```

and find/add/uncomment the lines and set these values:

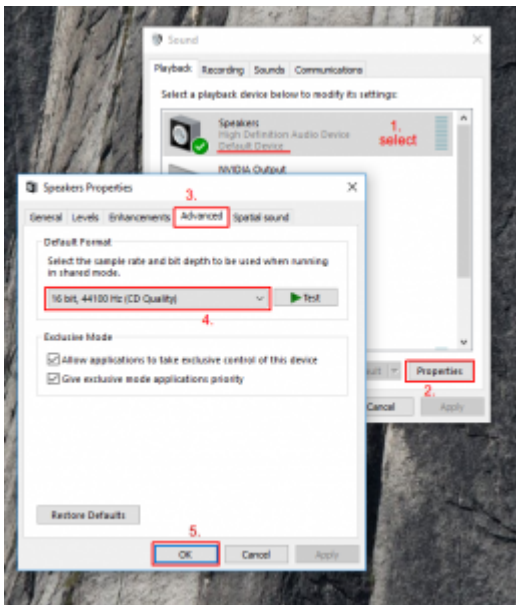
```
default-sample-rate = 44100
```

```
alternate-sample-rate = 48000
```

afterwards restart the pulse audio service via
`pulseaudio -k` or reboot.

Setting default sample rate in Windows

Set the frequency of your windows sound to 44100 Hz

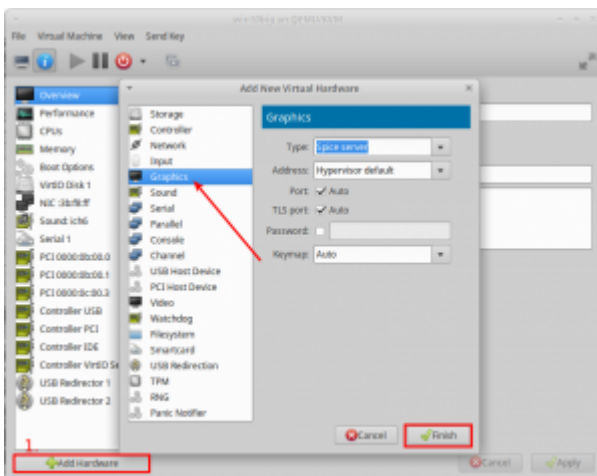


Set sample rate for default audio device in windows 10

Option 1 – use Spice Server

- (+) Easy to set up
- (+) Very good for first setup of the VM OS.
- (-) Needs VM window open in order to work
- (-) a little more delay than direct pulse audio version
- (-) Adds an extra display to the VM.

This one is pretty straight forward, open the info page of the VM on virt-manager and add a spice server.



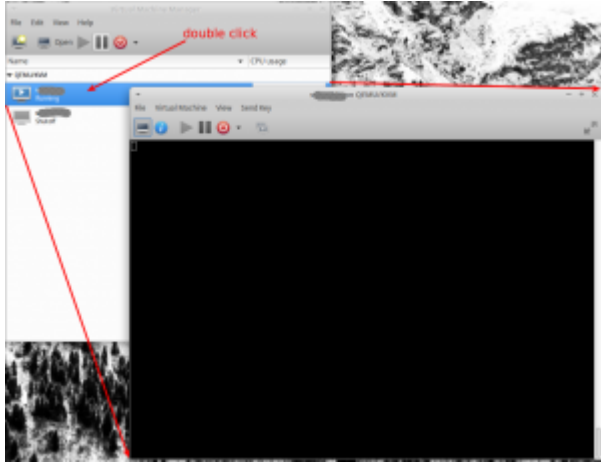
([https://mathiashueber.com/wp-](https://mathiashueber.com/wp-content/uploads/2018/02/Selection_012.png)

[content/uploads/2018/02/Selection_012.png](https://mathiashueber.com/wp-content/uploads/2018/02/Selection_012.png))

Add spice server via virt-manager GUI

Adding the spice server adds actually several devices to the VM config, one for example is a display.

In order to hear sound, I have to open the VM while it is running (I hope the sentence make sense 😊). Basically it is double clicking on the running VM in the virt-manager list.



([https://mathiashueber.com/wp-](https://mathiashueber.com/wp-content/uploads/2018/02/Selection_013.png)

[content/uploads/2018/02/Selection_013.png](https://mathiashueber.com/wp-content/uploads/2018/02/Selection_013.png))

Open VM spice server window while VM is running

Option 2 – use Pulse Audio

- (+) Less sound delay than the spice server version
- (+) Less overhead once running
- (+) Nicer Integration
- (-) Complicated to get it working
- (-) Lots of users complain about bad sound quality

Pulse Audio with QEMU 4.2 (and above)

Edit the virtual machine config via

```
virsh edit <your vm name>
```

e.g. `virsh edit windows10`

Make sure the very first line of the file does read:

```
<domain type='kvm' xmlns:qemu='http://libvirt.org/schemas/domain/qemu/1.0'>
```

Instead of

```
<domain type='kvm'>
```

Check at the bottom of your config if a line `<qemu:commandline>` exists. If yes make sure to add these options:

```
<qemu:arg value='-device' />
<qemu:arg value='ich9-intel-hda,bus=pcie.0,addr=0x1b' />
<qemu:arg value='-device' />
<qemu:arg value='hda-micro,audiodev=hda' />
<qemu:arg value='-audiodev' />
<qemu:arg value='pa,id=hda,server=unix:/run/user/1000/pulse/native' />
</qemu:commandline>
```

In case `<qemu:commandline>` is missing, find the line which ends with `</devices>`

and add the following block afterwards:

```
<qemu:commandline>
  <qemu:arg value='-device' />
  <qemu:arg value='ich9-intel-hda,bus=pcie.0,addr=0x1b' />
  <qemu:arg value='-device' />
  <qemu:arg value='hda-micro,audiodev=hda' />
  <qemu:arg value='-audiodev' />
  <qemu:arg value='pa,id=hda,server=unix:/run/user/1000/pulse/native' />
</qemu:commandline>
```

The 1000 in

```
server=unix:/run/user/1000/pulse/native
```

represents your user-id, 1000 is the default (one user) Id.

”Attention!

Make sure `<devices>` and `<qemu:commandline>` have the same indent.

” Remark!

The `<qemu:commandLine>` part can hold further options, simply append the sound options if others were already present.

Open the apparmor libvirt abstractions file via

```
sudo nano /etc/apparmor.d/abstractions/libvirt-qemu
```

Append the following lines

```
/etc/pulse/client.conf.d/ r,  
/etc/pulse/client.conf.d/* r,  
/run/user/1000/pulse/native rw,  
/home/your-username/.config/pulse/* r,  
/home/your-username /.config/pulse/cookie k,
```

Replace **your-username** with your username.

→ reboot your system.

Settings in Virt-manager

Make sure no further audio devices are added by virt-manager – remove all audio devices from the virtual hardware details bar (left side in VM info view).

Troubleshooting

If you have no sound at all, run `pax11publish` and check if a server with name `/run/user/1000/pulse/native` is available.

Pulse Audio with QEMU 4.1 (and below)

Edit the virtual machine config via

```
virsh edit <your vm name>
```

e.g. `virsh edit windows10`

Make sure the very first line of the file does read:

```
<domain type='kvm' xmlns:qemu='http://libvirt.org/schemas/domain/qemu/1.0'>
```

instead of

```
<domain type='kvm'>
```

Check at the bottom of your config if a line `<qemu:commandline>` exists. If yes make sure to add these options:

```
<qemu:env name='QEMU_AUDIO_DRV' value='pa' />
<qemu:env name='QEMU_PA_SAMPLES' value='8192' />
<qemu:env name='QEMU_AUDIO_TIMER_PERIOD' value='99' />
<qemu:env name='QEMU_PA_SERVER' value='/run/user/1000/pulse/native' />
```

In case `<qemu:commandline>` is missing, find the line which ends with `</devices>`

and add the following block afterwards:

```
<qemu:commandline>
  <qemu:env name='QEMU_AUDIO_DRV' value='pa' />
  <qemu:env name='QEMU_PA_SAMPLES' value='8192' />
  <qemu:env name='QEMU_AUDIO_TIMER_PERIOD' value='99' />
  <qemu:env name='QEMU_PA_SERVER' value='/run/user/1000/pulse/native' />
</qemu:commandline>
```

The 1000 in `/run/user/1000/pulse/native` represents your user-id. 1000 is the default (one user) user id.

”Attention!

After adding the enlightments, I had no internet on one Windows 10 guest (also not every time).

After some digging, I found a [reddit post](https://old.reddit.com/r/VFIO/comments/bsxclm/network_and_shutdownrestart_issues_with_windows/)

*(https://old.reddit.com/r/VFIO/comments/bsxclm/network_and_shutdownrestart_issues_with_windows/) with the proposed “solution” to change the **model type** of the network bridge interface in the libvirt config from “virtio” to “virtio-net-pci”.*

” *Remark!*
Lorem Ipsum!

Extract from my config file

Settings in Virt-manager

As said before, add a sound “ich6” device to your VM.

” *Attention!*
Do not use 6ch/9ch audio devices in Windows 10 1903 virtual machines. It creates awful stuttering and performance loss. Use ac97 audio devices instead.

Troubleshooting

If the sound distortions occur, make sure host and guest run at the same sample rate. You can also play around with `QEMU_PA_SAMPLES` and `QEMU_AUDIO_TIMER_PERIOD` values (see the reddit post(s) from the sources for further information).

Option 3 – use Hardware

(+) Very easy

(-) Costs money

I haven’t tried this, but it should be easily possible to add a USB sound card to the guest and run the output back into your host sound card via a real audio wire (connecting guest line-out with host line-in).
